Deep Learning Workshop

Setup Guide · Git Workflow · Environment

INSEA Innovation Edge · 2025

About this document

Send this guide to participants **2 days before the workshop**. It covers everything needed to arrive ready to code — Python, Git, GitHub, and the workshop repository. Follow each step in order. If anything fails, contact the organiser before the session.

Prerequisites Checklist

Complete all items below **before the day of the workshop**:

- [ ] Python 3.9 or higher installed
- [ ] `pip` working in your terminal
- [ ] Git installed and configured with your name and email
- [ ] A GitHub account created (free)
- [ ] The workshop repository cloned on your machine
- [ ] Dependencies installed via `pip install -r requirements.txt`
- [ ] Jupyter notebook or Google Colab confirmed working
- [ ] You can run `import torch` in a notebook without errors

Python Setup

Install Python

Download Python 3.10 or 3.11 from python.org/downloads. During installation on Windows, check **"Add Python to PATH"**.

Verify the installation by opening a terminal and running:

```
$ python -version
```

You should see `Python 3.x.x`. If the command is not found, restart your terminal or check that Python is in your system PATH.

Verify pip

`pip` is Python's package manager and is included with Python 3.4+. Confirm it works:

```
$ pip -version
```

> On some systems you may need to use `python3` and `pip3` instead of `python` and `pip`. Both work identically.

Git Setup

Install Git

- **Windows** — download from git-scm.com and run the installer (keep all default options).
- **macOS** — run `xcode-select -install` in the terminal, or install via Homebrew: `brew install git`.
- **Linux (Ubuntu/Debian)** — `sudo apt install git`

Verify:

```
$ git -version
```

Configure Git (one-time setup)

Tell Git who you are. These details appear in every commit you make:

```
$ git config -global user.name "Firstname Lastname"
```

```
$ git config -global user.email "you@example.com"
```

Confirm the configuration:

```
$ git config -list
```

GitHub Account Setup

1 Create a free account at github.com

Use a professional username — this will appear on your portfolio.

2 Verify your email address

GitHub sends a verification link to the email you registered with. Click it before proceeding.

3 (Optional but recommended) Set up SSH authentication

SSH lets you push code without typing your password every time.

```
$ ssh-keygen -t ed25519 -C "you@example.com"
```

Press Enter to accept the default file location, then press Enter twice (no passphrase needed for a workshop machine).

Copy the public key to your clipboard:

```
$ cat  /.ssh/id_ed25519.pub
```

Go to **GitHub → Settings → SSH and GPG keys → New SSH key**, paste the key and save.

Test the connection:

```
$ ssh -T git@github.com
```

You should see: *Hi username! You've successfully authenticated.*

Clone the Workshop Repository

1 Navigate to a folder where you want to store the project

```
$ cd  /Documents
```

2 Clone the repository

```
$ git clone https://github.com/YOUR-USERNAME/deep-learning-workshop.git
```

3 Enter the project folder

```
$ cd deep-learning-workshop
```

4 Confirm the contents

```
$ ls
```

You should see: `notebooks/` `slides/` `requirements.txt` `README.md` `.gitignore`

Install Dependencies

All required Python packages are listed in `requirements.txt`. Install them with one command:

```
$ pip install -r requirements.txt
```

This installs: `torch` · `torchvision` · `transformers` · `datasets` · `jupyter` · `matplotlib` · `numpy`

Verify PyTorch:

```
$ python -c "import torch; print(torch.__version__)"
```

Verify Transformers:

```
$ python -c "import transformers; print(transformers.__version__)"
```

Using Google Colab instead?

If you prefer not to install anything locally, all notebooks are ready to run on Google Colab with a free GPU. Open Colab, choose **File → Open notebook → GitHub**, paste the repository URL, and select the notebook you want. Run the first cell to install dependencies inside Colab.

Git Workflow During the Workshop

Create your personal branch

At the start of the session, create a branch with your name. This keeps your work separate from everyone else's:

```
$ git checkout -b workshop/firstname-lastname
```

Example: `git checkout -b workshop/sara-alami`

Verify you are on the right branch:

```
$ git branch
```

The active branch is marked with `*`.

Commit after each notebook

After completing each notebook, save your work with three commands:

```
$ git add notebooks/
```

```
$ git commit -m "feat:  math exercises done"
```

```
$ git push origin workshop/firstname-lastname
```

Use these commit messages at each checkpoint:

Message	When to use
`feat: math exercises done`	After notebook 01
`feat: MNIST classifier done`	After notebook 02
`feat: fine-tune BERT done`	After notebook 03

Open a Pull Request at the end

When the session ends, open a Pull Request on GitHub to share your work:

1. Go to github.com and open the workshop repo.
2. Click **Pull requests** → **New pull request**.
3. Set **base** to `main` and **compare** to your branch.
4. Set the title to: `Workshop - Firstname Lastname`.
5. Click **Create pull request**.

The instructor will review and leave feedback directly on your PR.

Git Quick-Reference Cheatsheet

Repository

`git clone <url>`	Download a remote repo locally
`git status`	See changed / untracked files
`git log -oneline`	View commit history (compact)

Branches

`git branch`	List all local branches
`git checkout -b <name>`	Create and switch to a new branch
`git checkout <name>`	Switch to an existing branch

Saving work

`git add <file>`	Stage a specific file
`git add .`	Stage all changed files
`git commit -m "message"`	Save staged changes with a message
`git push origin <branch>`	Upload branch to GitHub

Syncing

```
git pull        Download latest changes from remote
git fetch       Fetch without merging
git diff        See what changed since last commit
```

Troubleshooting

Problem	Solution
`python` **not found**	Restart terminal. On Mac/Linux try `python3`.
`pip install` **fails**	Run `pip install -upgrade pip` first, then retry.
`import torch` **error**	Reinstall: `pip install torch torchvision`.
`git push` **rejected**	Run `git pull` first to sync, then push again.
SSH key not working	Use HTTPS clone URL instead of SSH.
Notebook won't open	Run `jupyter notebook` in the project folder.

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Questions before the workshop? Reach out to the organiser.